Internet Crimes and Cyber Security

Week: 06 **Session: 26**

Topic :Fake News Analysis : Social media Crimes

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Understanding Social Media Crimes and Fake News

5. Understand fake news mechanisms

Fake news" refers to false or misleading information that mimics the format of legitimate news. While sometimes used to describe accidental errors, it primarily refers to **fabricated content** designed to manipulate public opinion, generate advertising revenue, or serve a political agenda. To better understand this ecosystem, experts often categorize "fake news" into three distinct types of "information disorder":

•**Disinformation:** Content that is intentionally false and created to cause harm or profit.
•**Misinformation:** False information shared by someone who believes it is true and has no intent to deceive.
•**Malinformation:** Genuine, true information that is shared out of context with the specific intent to cause harm (e.g., leaking private details for harassment)

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Understanding fake news Propagation

AI-Generated Persona Networks: Millions of **synthetic identities** are deployed, featuring AI-generated profile photos, deepfake voices, and unique histories to mimic credible journalists or activists.

Eg : Generate a Random Name - Fake Name Generator
Random Person/Name Generator | User Identity, Account and Profile Generator

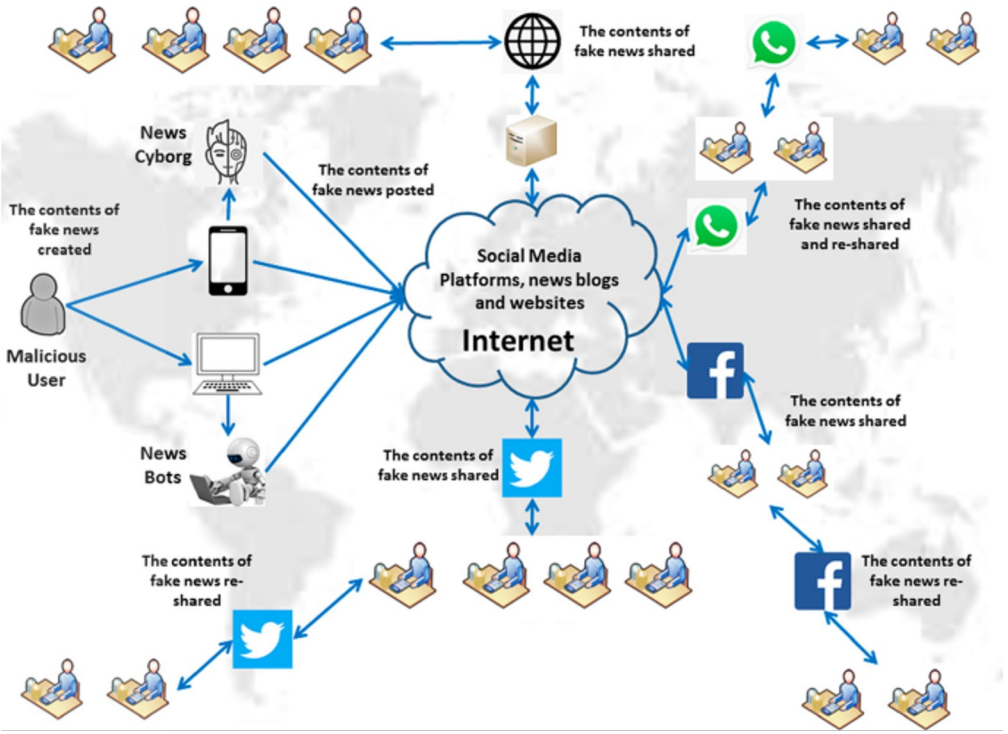
Automated Content Farms: Generative AI is used to produce mass quantities of low-cost propaganda. The number of AI-driven "fake news" sites has reportedly grown tenfold recently, often evading traditional detection by mimicking legitimate news styles.

LOVO: Free AI Voice Generator & Text to Speech

Microtargeting and Personalized Propaganda: AI-driven tools analyze psychological profiles to tailor disinformation to individual vulnerabilities, significantly increasing its persuasive power.

Fighting Disinformation Online A Database of Web Tools

Narrative Attack Automation: AI "agents" can now automate entire campaigns.



6. Tools for fake news detection

These tools use machine learning and natural language processing (NLP) to analyze content, sources, and spread patterns, often in real-time.

Logically AI: A mobile app and browser extension that combines AI with human fact-checkers to verify facts and images, monitoring vast numbers of domains and platforms in real-time.

The Factual: This tool scores news content daily (from 0 to 100) based on source quality, author expertise, and language objectivity. It is available as a browser extension, app, and newsletter.

Botometer: A web-based program that analyzes Twitter accounts to determine the likelihood of a profile being an automated bot, helping identify coordinated information manipulation.

Hoaxy: A tool from Indiana University that visualizes how information and fact-checking links spread online, helping researchers understand

Using Human-Driven Fact-Checking

Reverse Image Search (Google, TinEye, Duplichecker): Essential tools for uploading an image to find its original source, creation date, and any other instances online, which helps in identifying manipulated visuals.

InVid (Video Verification Plugin): A browser plugin toolkit designed for journalists to get contextual information and metadata about videos for verification.

Forensically: A web-based collection of tools for digital image forensics, including features like clone detection and error level analysis to spot image tampering.

Tools and techniques to Counter Cyber Attacks

6. AI Tools for fake news detection

1. Deepware Scanner

Deepware can scan video content and spot deepfake manipulation. It uses AI models trained on many fake and real videos to find inconsistencies in facial expressions, blinking, and lip movement.

2. Sensity AI

Sensity gives enterprise-level protection against deepfake attacks. It allows platform API access and supports manipulated video, audio, and images. It also sends alerts when suspicious content is found.

3. Hive Moderation

Hive AI can spot deepfake videos on social media platforms and streaming services, helping platforms remove harmful content quickly

4. ScamAdviser AI

ScamAdviser helps you spot fake stuff online, like dodgy websites and phishy emails.

5. Lookout Mobile Security

Lookout's app uses AI to protect smartphones from scam links, fake apps, and risky Wi-Fi networks. It provides immediate protection and blocks harm.

6. Multi-format Detection: Good tools can analyze videos, audio files, images, and texts.

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Concepts to be covered

1. Define social media crimes
2. Identify major Social Media crime categories
3. Introduction to SOCMINT (Social Media Intelligence) for Investigation
4. SOCMINT Case study using Maltego
5. Understanding fake news mechanisms and propagation
6. AI Tools for fake news detection

Thank You